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EDUCATION

Ph.D., Electrical and Computer Engineering, University of Florida, Gainesville, Florida 5/2014
“Learning and exploiting recurrent patterns in neural data”, Advisor: Jose C. Principe
B.S., Computer Engineering, University of Nebraska–Lincoln (Omaha campus) 5/2009
Highest Distinction, 2nd Major: Mathematics, Minor: Computer Science (GPA: 4.0/4.0)

EXPERIENCE

Assistant Professor University of Delaware, Newark, Delaware 12/2018–Present
Electrical and Computer Engineering; Computer and Information Sciences
Data Science Institute (Resident Faculty)
Research Fellow University of Manchester, United Kingdom 3/2017–10/2018
School of Computer Science, Mentor: Sophia Ananiadou
Research Associate University of Liverpool, United Kingdom 6/2014–2/2017
School of Electrical Engineering, Electronics & Computer Science, Mentor: John Y. Goulermas
Graduate Research Assistant University of Florida 5/2010–5/2014
Department of Electrical and Computer Engineering, Advisor: Jose C. Principe
Research Assistant University of Nebraska–Lincoln (Omaha Campus) Summer 2008/2009
Department of Computer and Electronics Engineering, Research Advisor: Hamid Sharif
Electronics Engineer Cenatmed, LLC, Omaha, NE 4/2008–7/2009
IT Operations Intern Union Pacific Railroad, Omaha, NE 8/2006–8/2008

HONORS AND AWARDS

International and National
– Gold Reviewer, International Conference on Machine Learning 2026
– Top Reviewer, Neural Information Processing Systems (NeurIPS) 2025
– IEEE Senior Member 2025
– Highlighted Reviewer of ICLR 2022 (top 8.7%=543/6207) 2022
– Top 5 List, Neural Engineering Community Award, IEEE EMBS NER Conf. 2021
– Top 200 Reviewer, Neural Information Processing Systems (NeurIPS) 2018
– Finalist, IEEE EMBS Conference Student Paper Competition 2013
– Honorable Mention, NSF Graduate Research Fellowship 2009, 2010, 2011
University of Delaware, Graduate College
– Most Engaged Advisor/Advisee Award, NRT MIDAS Traineeship Program 2024
University of Delaware, College of Engineering
– Faculty Award for Excellence in Service and Community Engagement 2022
University of Florida
– Honorable Mention, Outstanding Service, Graduate Student Council 2014
– Graduate School Fellowship 2009–2013
University of Nebraska
– Dean’s Award, College of Engineering 2009
– Outstanding Senior, Computer and Electronics Engineering 2009
– Undergraduate Major Honoree, Computer and Electronics Engineering 2009
– 1st Place Senior Thesis Design Team, Computer and Electronics Engineering 2009
– James Earl Mathematics Scholarship, Math. Dept., U. Nebraska Omaha 2008–2009
– Tau Beta Pi Distinguished Freshman Award 2006

- Walter Scott Jr. Scholarship, Peter Kiewit Institute 2005–2009
- University of Nebraska Regents Scholarship 2005–2009

Research

Publications Key: *—graduate advisee, †—undergrad. advisee, ‡—mentored visiting scholar.

JOURNAL ARTICLES, PEER-REVIEWED

- J28. M. I. Cano Achuri[‡], M. K. Lara, K. Abed Rabbo, B. T. Wilson, A. Meek*, J. M. Mahoney, A. E. Hernan, and A. J. Brockmeier, “Predicting seizure-model genotypes of mice from EEG waveforms,” *Journal of Neural Engineering*, Special Issue on Epilepsy and Neural Engineering, **23**(3):036016, 2026.
- J27. W. F. Fortino, F. B. Bianco, P. Protopapas, D. Muthukrishna, and A. Brockmeier, “ABC-SN: Attention Based Classifier for Supernova Spectra,” *The Astrophysical Journal*, **1000**(1):14, 2026.
- J26. S. Bhowmik M. Anandakrishnan, L. Klein, C. Arighi, M. Gioioso, C. Wu, A. Brockmeier, K. Vijay-Shanker, and C. Chen, “Integrating Text Mining and Knowledge Graph to Enhance Biopharmaceutical Process Optimization,” *PLOS ONE*, **21**(1): e0339197, 2026.
- J25. H. Baker*, M. S. Emigh, and A. J. Brockmeier, “Weakly Supervised Object Segmentation by Background Conditional Divergence,” *Transactions on Machine Learning Research (TMLR)* **2JJZhGvMW**, 2025.
- J24. K. M. Holton* , S. Y. Chan, A. J. Brockmeier, and M.-H. Hall, “Latent Growth Models of Longitudinal Changes in Functional Connectivity during Early Stage Psychosis,” *Neuroinformatics*, **23**(3):43, 2025.
- J23. C. C. Claros*, M. N. Anderson, W. Qian, A. J. Brockmeier, and T. A. Buckley, “A Machine Learning Model for Post-Concussion Musculoskeletal Injury Risk in Collegiate Athletes,” *Sports Medicine*, **55**(8):1971–1982, 2025.
- J22. K. M. Holton* , A. Higgins, A. J. Brockmeier, and M.-H. Hall, “Uncovering key predictive channels and clinical variables in the gamma band auditory steady-state response in early-stage psychosis: a longitudinal study” *Acta Neuropsychiatrica*, Cambridge University Press, **37**(e1), 2025.
- J21. C. C. Claros-Olivares* , R. G. Clements, G. McIlvain, C. L. Johnson, and A. J. Brockmeier, “MRI-based whole-brain elastography and volumetric measurements to predict brain age” *Biology Methods and Protocols*, Oxford University Press, **bpae086**, 2024.
- J20. M. N. Anderson, C. C. Claros*, W. Qian, A. Brockmeier, and T. A. Buckley, “Integrative Data Analysis to Identify Persistent Post-Concussion Deficits and Subsequent Musculoskeletal Injury Risk: Project Structure and Methods,” *BMJ Open Sport & Exercise Medicine*, **10**(1), 2024.
- J19. B. Riaz*, Y. Karahan*, and A. J. Brockmeier, “Partial Optimal Transport for Support Subset Selection,” *Transactions on Machine Learning Research (TMLR)*, **75CcopPxIr**, 2023.
- J18. K. Holton*, S. Y. Chan, A. J. Brockmeier, D. Öngür, and M.-H. Hall, “Exploring the influence of functional architecture on cortical thickness networks in early psychosis – A longitudinal study,” *NeuroImage*, **274**(120127), 2023.
- J17. E. N. Hamulyák, A. J. Brockmeier, J. D. Killas, S. Ananiadou, S. Middeldorp, and A. M. Leroi, “Women’s health in *The BMJ*: a data science history,” *BMJ Open*, **10**:e039759, 2020.
- J15. A. J. Brockmeier, M. Ju, P. Przybyła, and S. Ananiadou, “Improving reference prioritisation with PICO recognition,” *BMC Medical Informatics and Decision Making*, **19**(256), 2019.

- J16. X. Evangelopoulos, A. J. Brockmeier, T. Mu, J. Y. Goulermas, “Circular object arrangement using spherical embeddings,” *Pattern Recognition*, 103(107192), 2020.
- J14. P. Przybyła, A. J. Brockmeier, and S. Ananiadou, “Quantifying risk factors in medical reports with a context-aware linear model,” *Journal of the American Medical Informatics Association*, 26(6):537–546, 2019.
- J13. X. Evangelopoulos, A. J. Brockmeier, T. Mu, J. Y. Goulermas, “Continuation methods for approximate large scale object sequencing,” *Machine Learning*, 108(4):595–626, 2019.
- J12. P. Przybyła, A. J. Brockmeier, G. Kontonatsios, M.-A. Le Pogam, J. McNaught, E. von Elm, K. Nolan, and S. Ananiadou, “Prioritising references for systematic reviews with Robot-Analyst: A user study,” *Research Synthesis Methods*, 9(3):470–488, 2018.
- J11. A. J. Brockmeier, T. Mu, S. Ananiadou, and J. Y. Goulermas, “Self-tuned descriptive document clustering using a predictive network,” *IEEE Transactions on Knowledge and Data Engineering*, 30(10):1929–1942, 2018.
- J10. A. J. Brockmeier, T. Mu, S. Ananiadou, and J. Y. Goulermas, “Quantifying the informativeness of similarity measurements,” *Journal of Machine Learning Research*, 18(76):1–61, 2017.
- J9. G. Kontonatsios, A. J. Brockmeier, P. Przybyła, J. McNaught, T. Mu, J. Y. Goulermas, and S. Ananiadou, “A semi-supervised approach using label propagation to support citation screening,” *Journal of Biomedical Informatics*, 72:67–76, 2017.
- J8. J. S. Choi, A. J. Brockmeier, D. McNiel, L. von Kraus, J. C. Principe, and J. T. Francis, “Eliciting naturalistic cortical responses with a sensory prosthesis via optimized microstimulation,” *Journal of Neural Engineering*, 13(5):056007, 2016.
- J7. A. J. Brockmeier and J. C. Principe, “Learning recurrent waveforms within EEGs,” *IEEE Transactions on Biomedical Engineering*, 63(1):43–54, 2016.
- J6. M. S. Emigh, E. G. Krininger, A. J. Brockmeier, J. C. Principe, and P. M. Pardalos, “Reinforcement learning in video games using nearest neighbor interpolation and metric learning,” *IEEE Transactions on Computational Intelligence and AI in Games*, 8(1):56–66, 2016.
- J5. J. C. Principe and A. J. Brockmeier, “Representing and decomposing neural potential signals,” *Current Opinion in Neurobiology*, 31:13–17, 2015.
- J4. A. J. Brockmeier, J. S. Choi, E. G. Krininger, J. T. Francis, and J. C. Principe, “Neural decoding with kernel-based metric learning,” *Neural Computation*, 26(6):1080–1107, 2014.
- J3. L. Li, A. J. Brockmeier, J. S. Choi, J. T. Francis, J. C. Sanchez, and J. C. Principe, “A tensor-product-kernel framework for multiscale neural activity decoding and control,” *Computational Intelligence and Neuroscience*, Article ID 87016, 2014.
- J2. L. Li, I. M. Park, A. Brockmeier, B. Chen, S. Seth, J. T. Francis, J. C. Sanchez, and J. C. Principe, “Adaptive inverse control of neural spatiotemporal spike patterns with a reproducing kernel Hilbert space (RKHS) framework,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 21(4):532–543, 2013.
- J1. J. S. Choi, M. M. DiStasio, A. J. Brockmeier, and J. T. Francis, “An electric field model for prediction of somatosensory (S1) cortical field potentials induced by ventral posterior lateral (VPL) thalamic microstimulation,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 20(2):161–169, 2012.

CONFERENCE PAPERS, PEER-REVIEWED

- C29. H. Baker* and A. J. Brockmeier, “Patch2Loc: Learning to Localize Patches for Unsupervised Brain Lesion Detection,” *The 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, **Spotlight (2.5%)**, <https://openreview.net/forum?id=F5nwzHc5Nv>, 2026.

- C28. H. Baker* and A. J. Brockmeier, “Efficient Many-to-Many MRI Modality Translation via a Latent-Conditioned Vector-Quantized Network,” *12th Int. IEEE/EMBS Conf. Neural Engineering (NER)*, San Diego, CA, USA, 2025, pp. 1166-1171, doi:10.1109/NER61569.2025.11588989.
- C27. J. K. Hoyos-Osorio, O. Skean, A. J. Brockmeier, and L. G. Sanchez Giraldo, “Representation Jensen-Renyi Divergence,” *IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2022.
- C26. H. Baker* and A. J. Brockmeier, “Exploring latent networks in resting-state fMRI using voxel-to-voxel causal modeling feature selection”, *Machine Learning for Health (ML4H) - Extended Abstract*, 2021.
- C25. K. Holton*, S. Y. Chan, A. J. Brockmeier, D. Öngür, and M-H. Hall “Exploring the influences of functional connectivity architecture on cortical thickness networks in patients with early psychosis”, *Machine Learning for Health (ML4H) - Extended Abstract* , 2021. (Presented poster, but opted out of archival list due to acceptance of journal paper.)
- C24. C. H. Mendoza-Cardenas* and A. J. Brockmeier, “Shift-invariant waveform learning on epileptic ECoG”, *IEEE Engineering in Medicine and Biology Society (EMBC)*, 2021.
- C23. C. H. Mendoza-Cardenas* and A. J. Brockmeier, “Searching for waveforms on spatially-filtered epileptic ECoG”, *Int. IEEE/EMBS Conf. Neural Engineering (NER)*, 2021.
- C22. H. Baker* and A. J. Brockmeier, “Local and sparse linear causal models for fMRI resting-state signals”, *Int. IEEE/EMBS Conf. Neural Engineering (NER)*, 2021.
- C21. X. Evangelopoulos, A. J. Brockmeier, T. Mu, and J. Y. Goulermas, “A graduated non-convexity relaxation for large scale seriation,” in *SIAM Int. Conf. Data Mining (SDM)*, 2017.
- C20. M. Sato, A. J. Brockmeier, G. Kontonatsios, T. Mu, J. Y. Goulermas, J. Tsujii, and S. Ananiadou, “Distributed document and phrase co-embeddings for descriptive clustering,” in *European Chapter of the Association for Computational Linguistics (EACL)*, 2017.
- C19. A. J. Brockmeier and J. C. Principe, “Explicit versus implicit source estimation for blind multiple input single output system identification,” in *IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2015.
- C18. E. Santana, A. J. Brockmeier, and J. C. Principe, “Joint optimization of algorithmic suites for EEG analysis,” in *IEEE Engineering in Medicine and Biology Society (EMBC)*, 2014.
- C17. A. J. Brockmeier, E. Santanna, L. Sanchez Giraldo, and J. Principe, “Projentropy: Using entropy to optimize spatial projections,” in *IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2014.
- C16. A. J. Brockmeier, L. G. Giraldo, J. S. Choi, J. T. Francis, and J. C. Principe, “Learning multiscale neural metrics via entropy minimization,” in *Int. IEEE/EMBS Conf. Neural Engineering (NER)*, 2013.
- C15. A. J. Brockmeier, L. G. Sanchez Giraldo, M. S. Emigh, J. Bae, J. S. Choi, J. T. Francis, and J. C. Principe, “Information-theoretic metric learning: 2-D linear projections of neural data for visualization,” in *IEEE Engineering in Medicine and Biology Society (EMBC)*, 2013.
- C14. A. J. Brockmeier, J. C. Principe, A. H. Phan, and A. Cichocki, “A greedy algorithm for model selection of tensor decompositions,” in *IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2013.
- C13. A.-H. Phan, A. Cichocki, P. Tichavsky, G. Luta, and A. Brockmeier, “Tensor completion through multiple Kronecker product decomposition,” in *IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2013.
- C12. A. J. Brockmeier, M. K. Hazrati, W. J. Freeman, and J. C. Principe, “Locating spatial patterns of waveforms during sensory perception in scalp EEG,” in *IEEE Engineering in Medicine and Biology Society (EMBC)*, 2012.

- C11. A. J. Brockmeier, J. S. Choi, M. M. Emigh, J. T. Francis, and J. C. Principe, “Subspace matching thalamic microstimulation to tactile evoked potentials in rat somatosensory cortex,” in *IEEE Engineering in Medicine and Biology Society (EMBC)*, 2012.
- C10. B. H. Fadlallah, A. J. Brockmeier, S. Seth, L. Li, A. Keil, and J. C. Principe, “An association framework to analyze dependence structure in time series,” in *IEEE Engineering in Medicine and Biology Society (EMBC)*, 2012.
- C9. A. J. Brockmeier, B. Mahmoudi, J. C. Sanchez, and J. C. Principe, “Efficient temporal decomposition of local field potentials,” in *IEEE Int. Work. Machine Learning for Signal Processing (MLSP)*, 2011.
- C8. A. J. Brockmeier, J. S. Choi, M. M. DiStasio, J. T. Francis, and J. C. Principe, “Optimizing microstimulation using a reinforcement learning framework,” in *IEEE Engineering in Medicine and Biology Society (EMBC)*, 2011.
- C7. S. Craciun, A. J. Brockmeier, A. D. George, H. Lam, and J. C. Principe, “An information-theoretic approach to motor action decoding with a reconfigurable parallel architecture,” in *IEEE Engineering in Medicine and Biology Society (EMBC)*, 2011.
- C6. S. Seth, A. J. Brockmeier, J. S. Choi, M. Semework, J. T. Francis, and J. C. Principe, “Evaluating dependence in spike train metric spaces,” in *Int. Joint Conf. Neural Networks (IJCNN)*, 2011.
- C5. S. Seth, A. J. Brockmeier, and J. C. Principe, “A metric approach toward point process divergence,” in *IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2011.
- C4. A. J. Brockmeier, E. G. Kriminger, J. C. Sanchez, and J. C. Principe, “Latent state visualization of neural firing rates,” in *Int. IEEE/EMBS Conf. Neural Engineering (NER)*, 2011.
- C3. L. Li, A. Brockmeier, J. T. Francis, J. C. Sanchez, and J. C. Principe, “An adaptive inverse controller for online somatosensory microstimulation optimization,” in *Int. IEEE/EMBS Conf. Neural Engineering (NER)*, 2011.
- C2. S. Seth, I. Park, A. Brockmeier, M. Semework, J. Choi, J. Francis, and J. Principe, “A novel family of non-parametric cumulative based divergences for point processes,” in *Advances in Neural Information Processing Systems (NIPS)*, 2010.
- C1. A. J. Brockmeier, I. Park, B. Mahmoudi, J. C. Sanchez, and J. C. Principe, “Spatio-temporal clustering of firing rates for neural state estimation,” in *IEEE Engineering in Medicine and Biology Society (EMBC)*, 2010.

WORKSHOP PAPERS (PEER-REVIEWED)

- W5. C. C. Claros-Olivares* and A. J. Brockmeier “Bounding Worst-Case Calibration Error in OOD Detection Under Distribution Shift,” Towards Trustworthy Predictions: Theory and Applications of Calibration for Modern AI (Calibration for Modern AI @ AISTATS 2026), <https://openreview.net/forum?id=aZnAoyzssI>, 2026.
- W4. A. Meek*, C. H. Mendoza-Cardenas*, and A. J. Brockmeier, “Convolutional Monge Mapping between EEG Datasets to Support Independent Component Labeling,” *NeurIPS 2025 Workshop on Learning from Time Series for Health (TS4H)* <https://openreview.net/pdf?id=phmVS0qNo5>, 2025.
- W3. C. H. Mendoza-Cardenas*, A. Meek*, and A. J. Brockmeier, “Labeling EEG Components with a Bag of Waveforms from Learned Dictionaries,” *ICLR 2023 Workshop on Time Series Representation Learning for Health*, 2023.
- W2. Y. Karahan*, B. Riaz*, and A. J. Brockmeier, “Kernel landmarks: An empirical statistical approach to detect covariate shift”, *Workshop on Distribution Shifts, 35th Conference on Neural Information Processing Systems (NeurIPS 2021)*.
- W1. A. J. Brockmeier, C. C. Claros Olivares*, M. S. Emigh, and L. G. Sanchez Giraldo “Identifying the instances associated with distribution shifts using the max-sliced Bures divergence”, *Workshop on Distribution Shifts, 35th Conference on Neural Information Processing Systems (NeurIPS 2021)*.

CONFERENCE PAPERS REFEREED VIA EXTENDED ABSTRACT

- E2. M. S. Emigh, H. Baker*, C. H. Mendoza-Cardenas*, and A. J. Brockmeier, “Weakly supervised automatic target masking for synthetic aperture sonar,” *5th International Conference on Synthetic Aperture in Sonar and Radar*, Lerici, Italy, 2023.
- E1. H. Phan*, M. J. Wardlaw, B. Kim, and A. J. Brockmeier, “Training a Machine Learning Model for Underwater Chemical Source Localization in Simulated Turbulent Flows,” in *OCEANS 2022*, Hampton Roads, Virginia, 2022.
- OTHER PRE-PRINTS O2. O. Skean, J. K. Hoyos Osorio, A. J. Brockmeier, and L. G. Sanchez Giraldo, “DiME: Maximizing Mutual Information by a Difference of Matrix-Based Entropies,” *arXiv*, <https://arxiv.org/pdf/2301.08164>.
- O1. A. J. Brockmeier, Y. Karahan*, C. C. Claros*, C. H. Mendoza-Cardenas*, M. S. Emigh, and L. G. Sanchez Giraldo, “Max-sliced Bures Distance for Interpreting Discrepancies,” https://openreview.net/forum?id=D2Fp_qheYu, 2021.
- BOOK CHAPTER A. J. Brockmeier and J. C. Príncipe, “Decoding algorithms for brain machine interfaces,” in *Neural Engineering*, Bin He, Ed. Springer, 2013, pp. 223–257. (peer-reviewed)
- PATENT U.S. Patent 10,531,806. J. Principe and A. J. Brockmeier, “Brain state advisory system and methods using calibrated metrics and optimal time-series decomposition,” 1/14/2020.

PH.D. DISSERTATION SUPERVISION (COMMITTEE CHAIR)

6. Bilal Riaz, Ph.D. in Electrical and Computer Engineering Fall 2025
“Applications of computational optimal transport in machine learning and signal processing”
5. Hassan Baker, Ph.D. in Electrical and Computer Engineering Fall 2025
“Improving learning under data scarcity constraints: application in brain MRI and natural images”
4. Yalin Liao, Ph.D. in Electrical and Computer Engineering Spring 2025
“Statistical divergences and density estimation for anomaly detection and generative modeling”
3. Yuksel Karahan, Ph.D. in Electrical and Computer Engineering Fall 2024
“Detecting distributional discrepancies using kernel landmarks”
2. Kristina M. Holton, Ph.D. in Bioinformatics Data Science Spring 2024
“Exploring early stage psychosis through multimodal approaches: a longitudinal study”
1. Carlos H. Mendoza-Cardenas, Ph.D. in Electrical and Computer Engineering Winter 2023
“Learning representative waveforms to analyze, summarize, and compare long-term neural recordings”

MASTER’S THESES SUPERVISION

3. Hau Van Phan, M.S. in Electrical and Computer Engineering Summer 2022
“Training a machine learning model for underwater chemical source localization in simulated turbulent flows”
2. Bilal Riaz, M.S. in Electrical and Computer Engineering Spring 2022
“On spectral clustering, informativeness and seriation”
1. C. Cesar Claros, M.S. in Electrical and Computer Engineering Summer 2020
“Synergistic human-machine prediction: Active error analysis and mitigation with Gaussian process regression”

ABSTRACTS

- A15. E. T. Mans[†], H. Baker* and A. J. Brockmeier, “Predicting the Spatial Origin of EEG Independent Components from their Spectral-Temporal Features’,” *Int. IEEE/EMBS Conf. Neural Engineering (NER’25)*, 11/2025.

- A14. C. Chen, S. Bhowmik, M. Anandkrishnan, L. Klein, C. Arighi, M. Gioioso, C. Wu, A. Brockmeier, and K. Vijay-Shanker, "Combining Text-Mining and Knowledge Graph Approach to Inform Glycan Profile Optimization in Biopharmaceutical Manufacturing," *9th Annual MidAtlantic Bioinformatics Conference*, Philadelphia, Pennsylvania, 10/2024.
- A13. T. A. Buckley, A. Brockmeier, Qian, C. Claros, and M. Anderson "Predictors of Post-Concussion Musculoskeletal Injury Severity: An Integrated Data Analysis Approach," *United Kingdom Collaborating Centre on Injury and Illness Prevention in Sport: International Olympic Committee Research Centre*, Edinburgh, UK, 7/2024.
- A12. M. S. Emigh, H. Baker, B. Riaz, J. L. Prater, J. J. Dale, Y. Li, and A. Brockmeier "Improving SAS automatic object recognition with sub-aperture imagery from circular SAS," *International Conference on Underwater Acoustics*, Bath, UK, 6/2024.
- A11. T. A. Buckley, M. N. Anderson, A. Brockmeier, and W. Qian "Developing a Post-Concussion Injury Prediction Model," *International Olympic Committee Conference: Prevention of Injury and Illness in Sport*, Monaco City, Monaco, 2/2024.
- A10. K. Holton, A. Higgins, A. J. Brockmeier, and M.-H. Hall "Uncovering Key Predictive Channels and Clinical Variables in the Gamma Band Auditory Steady-State Response in Early Stage Psychosis - a Longitudinal Study," in *11th International IEEE EMBS Conference on Neural Engineering (NER'23)*, Baltimore, Maryland, 4/2023.
- A9. H. Baker and A. J. Brockmeier "Patch2Loc: Learning Representations to Localize MRI Patches for Abnormality Detection," in *11th International IEEE EMBS Conference on Neural Engineering (NER'23)*, Baltimore, Maryland, 4/2023.
- A8. C. H. Mendoza-Cardenas, A. Meek, and A. J. Brockmeier, "Labeling EEG Components with a Bag of Waveforms from Learned Dictionaries," in *11th International IEEE EMBS Conference on Neural Engineering (NER'23)*, Baltimore, Maryland, 4/2023.
- A7. G. McIlvain, R. Clements, C. C. Claros Olivares*, A. J. Brockmeier, C. L. Johnson "Mechanical property-based brain age prediction using artificial neural networks", *World Congress of Biomechanics*, Taipei, Taiwan, 7/2022.
- A6. K. Holton*, S. Y. Chan, A. J. Brockmeier, D. Öngür, and M-H. Hall "Exploring the influences of functional connectivity architecture on cortical thickness networks in patients with early psychosis", *60th Annual Meeting of the American College of Neuropsychopharmacology*, San Juan, Puerto Rico, 12/2021.
5. K. Nolan, S. Ananiadou, P. Przybyła, A. J. Brockmeier, "RobotAnalyst: An online system to support citation screening in evidence reviewing," at *Global Evidence Summit*, Cape Town, South Africa, 9/2017.
4. S. Dura-Bernal, K. Li, A. J. Brockmeier, C. C. Kerr, S. A. Neymotin, J. C. Principe, J. T. Francis, and W. W. Lytton, "Modulation of virtual arm trajectories via microstimulation in a spiking model of sensorimotor cortex," at *23rd Ann. Computational Neuroscience Meeting: CNS*2014*, Québec City, Canada, 7/2014.
3. E. Kriminger, A. Brockmeier, L. Sanchez-Giraldo, and J. Principe. "Metric learning for invariant feature generation in reinforcement learning," at *Reinforcement Learning and Decision Making*, Princeton, New Jersey, 10/2013.
2. J. S. Choi, A. J. Brockmeier, M. Emigh, L. von Kraus, and J. T. Francis. "Optimizing multi-channel microstimulation pulse trains with a model-predictive controller," at *23rd Ann. Meeting of the Society for the Neural Control of Movement*, San Juan, Puerto Rico, 4/2013.
1. E. K. Anderson, A. J. Brockmeier, N. G. Reyero, D. S. Barber, and N. D. Denslow. "Developing and validating a novel method for selecting class-specific biomarkers in ecotoxicology: A case study using fathead minnow microarray data," at *31st Ann. National SETAC Conf.*, Portland, Oregon, 11/2010.

- INVITED TALKS
9. “Finding Diagnostically-relevant Patterns in Neural Signals & Brain Structure with Machine Learning,” DARWIN Symposium, University of Delaware, 3/12/2026.
 8. “Finding Diagnostically-relevant Patterns in Neural Signals & Brain Structure with Machine Learning,” Neuroscience @ Nemours, Nemours Children’s Health, Wilmington, DE, 3/11/2026.
 7. “Disentangled Learning Representations of Sonar Data for Target Recognition,” Unmanned Maritime Systems Technology, Program Review, Office of Naval Research, Georgia Tech Research Institute, Smyrna, GA, 2/11/2026.
 6. “Machine Learning for Neuroimaging & Risk Profiling,” AI4Health Industry Day, 1/23/2026.
 5. “Divergence Metrics for Data Set Alignment and Generative Models,” Unmanned Maritime Systems Technology, Program Review, Office of Naval Research, Georgia Tech Research Institute, Smyrna, GA, 1/28/2025.
 4. “NextGen of Data Science: Trends and Challenges,” NextGen Data Science Symposium, UD DSI Student Association and Fellows, University of Delaware, 12/7/2023.
 3. “Machine learning methods for deciphering diagnostic patterns in brain structure and function,” Interdisciplinary Neuroscience Graduate Program Seminar, University of Delaware, 5/10/2023.
 2. “Finding patterns within brain waves”, Moberg Analytics and Sparse Coding Lab, Drexel University, 8/13/2021.
 1. “Mini Report by a JSPS Alumnus,” *Japan Society for Promotion of Science (JSPS) Fellowship Info Session*, University of Delaware’s Institute for Global Studies; Office of International Students & Scholars, Newark, Delaware, 11/21/2019.
- SHORT TALKS & PANELS
11. Spotlight Talk, “Patch2Loc: Learning to Localize Patches for Unsupervised Brain Lesion Detection”, AISTATS, 5/4/2026.
 10. Panelist, “Preparing the ECE Workforce in the AI Era” 12/4/2026. Community Summit, Electrical and Computer Engineering Department Heads Association, 12/4/2025.
 9. Panelist, “Bioinformatics, Data Science, and AI: Driving collaborative team science in biomedicine, life sciences, and beyond”, Delaware Bioinformatic Data Science Symposium, 10/23/2025.
 8. AI4Health Industry Day, 1/31/2025.
 7. AI Symposium, UD College of Engineering, 9/25/2023.
 6. Army C5ISR, 8/10/2023.
 5. “Immerse Delaware”, Waters Corporation, 12/21/2021.
 4. Lockheed Martin–Advanced Technology Laboratories, 10/5/2021.
 3. Waters Corporation, 8/31/2021.
 2. Chemours Company, 6/31/2021.
 1. Panelist, “Breakout session: Data science and precision medicine,” 2019 Delaware IDEAs Symposium, 11/7/2019.
- MENTIONS IN MEDIA
8. “New research decodes epilepsy in brain waves” AAAS EurekAlert! 6/4/2026
 7. UDaily “A game-changing tool” 4/15/2025 AAAS EurekAlert!
 6. UDaily “How old is your brain?” 3/24/2025
 5. UDaily “AI at UD” 10/24/2024
 4. UD Magazine “It’s the end of the world as we know it” 10/20/2024
 3. *Delaware Business Times* “UD’s Data Science Institute Harnesses Data as a Tool for Good” 8/6/2024
 2. Graduate College News “Big Data = Big Solutions” 3/21/2024
 1. UDaily “Preventing further injury in athletes after concussion” 1/27/2024

CURRENT
RESEARCH
GRANTS

- **Delaware ACCEL Interdisciplinary Collaboration Grant** 1/2026–12/2027
“Uncovering Hidden EEG Biomarkers of Epilepsy Using Machine Learning”, Role: **Lead PI** (MPI: Amanda Hernan, Nemours Children’s Health) 1 research assistant for 20 months across 2 years (Total Direct: \$96,828)
- **Delaware Community Foundation** 7/2025–6/2030
“Expanding our Partnership: Steps Towards a Cure”, Role: **Co-I** (PI: Christopher Martens, University of Delaware). Supports 1 month effort and 1 research assistant for 5 years (Total: \$13,102,616)
- **Office of Naval Research** N00014-24-1-2259 3/2024–2/2027
“Disentangled learning representations of sonar data for target recognition and semantic compression”, Role: **Sole PI**. 1.5 months effort per year, supports 1 research assistant for 3 years (\$370,000)
- **National Science Foundation** NSF 2108841 8/1/2021–7/31/2026
“Detecting and studying light echoes in the era of Rubin and artificial intelligence”, Role: **Co-PI**. 0.5 months effort in years 1 and 2 (PI: Federica Bianco, University of Delaware) (Total: \$596,068)

As Senior Personnel for Education/Training Grants:

- **National Science Foundation** NSF 2336586 - Supplement 12/23/2025–12/31/2028
Presidential AI-Youth Challenge Competition DCL Supplement, Role: **Senior Personnel** (PI: Ken Barner, University of Delaware). (Total: \$25,000)
- **National Science Foundation** NSF 2125703 9/1/2021–8/31/2026
“NRT-HDR: Computing and Data Science Training for Materials Innovation, Discovery, Analytics”, Role: **Senior Personnel** (PI: Arthi Jayaraman, University of Delaware). 0.21 months effort in year 1 and 0.17 months in subsequent years (Total: \$2,999,011)
- **National Science Foundation** NSF 2123264 9/15/2021–8/31/2026
“Collaborative Research: HDR DSC: Delaware and Mid-Atlantic Data Science Corps”, Role: **Senior Personnel** (PI: Federica Bianco, University of Delaware). 0.5 and 0.25 months effort in years 1 and 2 (Total: \$1,500,000)

Intramural:

- **University of Delaware Research Foundation** 6/1/2022–11/30/2025
UDRF: “Mapping and decoding the brain’s activity during human-AI interaction” Role: **PI**. Supports 1 research assistant for 1 year and two undergraduate researchers (summer), and neuroimaging (\$30,000 + \$5,000 in REU support + \$15,000 match)

COMPLETED
RESEARCH
GRANTS

- **Delaware Community Foundation** 1/2024–6/2025
“Neural mechanisms of Alzheimer’s disease risk and onset from modifiable lifestyle factors”, Role: **Co-I** (PI: Christopher Martens, University of Delaware). Supports 1 month effort and 1 research assistant for 1 year (Total: \$575,000)
- **Office of Naval Research** N00014-21-1-2300 4/2021–10/2024
“Interpretable maximal discrepancies metrics for analyzing and improving generative models”, Role: **Sole PI**. 1.75 months effort per year, supports 1 research assistant for 3 years (\$346,941)
- **Waters Technologies Corporation** 4/2023–4/2024
“Text Mining and Information Retrieval and Extraction”, Role: **Co-PI** (PI: Vijay Shanker, University of Delaware). 0.3 months effort (Total: \$372,593)
- **National Institutes of Health** R21 NS122033-01A 9/3/2021–7/31/2023
“Integrative Data Analysis to Identify Persistent Post-Concussion Deficits and Subsequent Musculoskeletal Injury Risk”, Role: **Co-I** (PI: Thomas A. Buckley, University of Delaware), 0.85 months effort in years 1 and 2, supports 1 research assistant for 2 years (Total: \$419,798)
- **Office of Naval Research** 6/1/2020–5/31/2022
(Subaward from *Minority Serving Institutions Science, Technology, Engineering and Mathematics Research & Development Consortium*) “Development of Bio-Inspired Nano-

Sensors for Underwater Explosives and Hazardous Materials”

Role: **Co-PI** (PI: Bruce Kim, City College of New York). 0.75 months effort per year, supports 1 research assistant for 2 years (Subaward of \$170,659)

Intramural:

- **Unidel Foundation**, UD’s AI Center for Excellence Seed Grant 1/1/2024–6/30/2024 “Adaptive thermal emission design driven by AI” Role: **Co-PI** (PI: Xi Wang, University of Delaware). Supports 1 research assistant from Wang’s group (\$15,000)
- **Unidel Foundation**, UD’s AI Center for Excellence Seed Grant 1/1/2023–6/30/2023 “Predicting after-effects of exoskeleton-assisted gait training to inform human-in-the-loop control optimization” Role: **Co-PI** (PI: Fabrizio Sergi, University of Delaware). In-kind undergraduate research support.
- **University of Delaware Research Foundation** 11/1/2019–10/31/2021 UDRF–SI: “Advancing machine learning for neuroimaging through topology-aware signal processing” Role: **PI** (Senior Mentor: Gonzalo Arce, University of Delaware). Supports 1 research assistant for 1.5 year (\$30,000 + \$15,000 match)
- **Unidel Foundation**, UD’s Data Science Institute Mini-Grant 9/12/2019–5/31/2020 “Learning to predict systematic errors in machine learning models and alert an expert for improved synergistic performance” Role: **PI**. Supports master’s student hourly research (\$10,000)

Contributed to Project Description for Research Instrumentation:

- **National Science Foundation**: “MRI: Acquisition of a Big Data and High Performance Computing System to Catalyze Delaware Research and Education”, Role: provided use case (PI: Rudolf Eigenmann, University of Delaware), in-kind support of computing access on DARWIN (Total: \$1,399,992)

Prior to Joining University of Delaware:

- JSPS Postdoctoral Fellowship Program (Short-term) 2014 “Algorithms for learning distance metrics to understand neural processing in the brain,” [Proposal accepted, but submission withdrawn.] Proposed host: Masashi Sugiyama, Tokyo Institute of Technology, Japan.
- JSPS Summer Program, NSF East Asia & Pacific Summer Institutes Fellow 2012 ‘Signal processing techniques to separate and analyze brainwaves’, hosted by Andrzej Cichocki, RIKEN Brain Science Institute, Japan.

CURRENT	4. Austin J. Meek, Computer Science	Fall 2022–present
PH.D.	Daniel L. Chester Graduate Fellow, 2024–present	
STUDENTS	3. Zhi Li (co-advisor J. Garcia-Frias)	Fall 2022–present
	ECE First Year Fellowship, Fall 2022	
	2. Justin Labombard (co-advisor K. Barner)	Fall 2022–present
	Senior Trainee, NRT: Computing & Data Science Training for Materials Innovation, Discovery, AnalyticS (MIDAS), 2025–2026	
	ECE Research Day Poster Award (Signal Processing, Communication, & Control Area) 5/7/2024	
	Trainee, NRT: Computing & Data Science Training for Materials Innovation, Discovery, AnalyticS (MIDAS), 2022–2024	
	ECE First Year Fellowship, Fall 2022	
	1. Claudio Cesar Claros Olivares	Spring 2022–present
	Bendett Fellowship Award, 2023	
	ECE Research Day Poster Award (Signal Processing, Communication, & Control Area) 5/8/2024	
PAST GRADUATE RESEARCH STUDENTS	(* active collaboration on further publications, [†] change of advisor)	
	Alex Mulrooney (co-advised with D. Hong for one semester)	Fall 2025–Winter 2026 [†]
	ECE First Year Fellowship, Gore Fellowship, NSF Graduate Research Fellowship Program (applied Fall 2025). Subsequent employment: Ph.D. student advised by D. Hong.	

Hassan Baker, Ph.D., ECE Spring 2020–Fall 2025*
 Summer Doctoral Fellowship, Graduate College, Summer 2025
 President, DSI Student Association, 2022—2024
 Gore Fellowship, 2021
 Subsequent employment: Post-doctoral Researcher, University of Texas at San Antonio

Bilal Riaz, Ph.D., ECE Fall 2019–Fall 2025*
 HEC Scholarship, US-Pakistan Knowledge Corridor, Higher Education Committee, 2019–2024
 Subsequent employment: Post-doctoral Researcher, University of North Carolina at Charlotte

Yalin Liao, Ph.D., ECE Fall 2022[†]–Spring 2025*
 Subsequent employment: Post-doctoral Researcher, Moffitt Cancer Center

Yüksel Karahan, Ph.D., ECE Spring 2019[†]–Fall 2024*
 Subsequent employment: Assistant Professor, National University; Bloom Energy
 Past position: Data Science Fellow, Delaware Data Innovation Lab
 ECE Research Day Poster Award (Signal Processing, Communication, & Control Area) 5/4/2022

Kristina Holton, Ph.D., Bioinformatics Data Science Spring 2020–Spring 2024*
 Concurrent/subsequent employment: Bioinformatician, Harvard University & Harvard Medical School
 Best Poster Award, DSI's 2021 Delaware Data Science Symposium

Carlos H. Mendoza-Cardenas, Ph.D., ECE Winter 2019[†]–Winter 2023*
 Subsequent employment: Applied Scientist, Twitch

Hau Van Phan, M.S., ECE Spring 2021–Summer 2022
 Subsequent employment: Machine Learning Engineer, Qlik

UNDERGRADUATE RESEARCHERS (* continuing collaboration on publications)

Daniel England, INBRE Summer Scholars	Summer 2026
Maxwell Morrison, UD Summer Scholars Program	Summer 2026
Eric Mans, UD ECE REU	Summer 2025
Vance Steele, UD ECE REU	Summer 2024
David Cardenas (co-mentored by Dr. Federica Bianco), Data Science Corps	Summer 2024
Alex Mulrooney, UD Summer Scholars Program	Summer 2022 & 2023
Research Assistant	Fall 2023–Spring 2024*
Travis Deputy, UD ECE REU	Summer 2023
Evan Curtin, UD Summer Scholars Program	Summer 2021
Justin Labombard, UD Summer Scholars Program	Summer 2021
Thomas Pisklak, UD Summer Scholars Program	Summer 2021

VISITING SCHOLARS

Maria Isabel Cano Achuri, Universidad de Antioquia, Colombia	Summer–Fall 2023
Karen Andrea Fonseca, Universidad Industrial de Santander, Colombia	Summer 2022
Andres Nicolas Lopez, MSc., National University of Colombia, Colombia	Summer 2021
Edwin Salcedo, M.Sc., M.B.A., Bolivian Catholic University, Bolivia	Summer 2019
Jose Luis Falla, M.Sc., National University of Colombia, Bogotá, Colombia (co-mentored with Drs. S. Singh and J. Garcia-Frias)	Summer 2019

Teaching

COURSES	University of Delaware , Newark, Delaware, USA	
	<i>Instructor</i> , Large Scale Machine Learning (ELEG/FSAN/CISC 817) Fall semesters: 2019–2025 Developed and delivered as a new course serving as a second and capstone course in machine learning for graduate students. Covers computational and statistical scaling of machine learning algorithms and tasks from both theoretical and practical perspectives. Content revised annually to encompass emerging machine learning approaches including generative models, transformers, and self-supervised learning. Approved as 800-level CISC elective in 2021 and 2022. Crosslisted with CISC in 2023. Enrollment: 13–28 students.	
	<i>Instructor</i> , Signals and Systems (ELEG 305) Spring semesters: 2020–2022, 2024–2026 Undergraduate core requirement for Electrical Engineering, Computer Engineering, Cybersecurity Engineering, and GIScience and Environmental Data Analytics. Developed additional programming assignments. Performed ABET continual performance through documented initiation, assessment of student outcomes, and self-evaluation. Enrollment: 60–96 students. Facilitated an Honors Add-on discussion section with independent research in 2021, 2024–2026.	
	<i>Co-Instructor</i> , Computing & Data Sci. for Soft Materials Innovation & Discovery Spring 2023 (CHEG/ELEG/CISC/MSEG 867) Co-instructor for industry project centered course. Academic mentor for two computationally focused projects; lectures on machine learning for the industry-project focused course; and provided student feedback. Enrollment: 14 students.	
	<i>Co-Instructor</i> , Geospatial Data Science (GEOG/CISC/ELEG/SPAA 367) Spring 2023 Co-developed as a new, experimental interdisciplinary data science course. Covered geospatial data science and machine learning as one of the three modules. Enrollment: 19 students.	
	<i>Coordinator</i> , Signal Processing & Communications Seminar (ELEG 663) Spring 2022, 2023 Coordinated external speakers discussing research in machine learning, signal processing, brain-machine interfaces, and information theory.	
	<i>Instructor</i> , Independent Study (ELEG 466) Summer 2021 Collected material, weekly discussions, and projects focused on “Adaptive Filtering as an Introduction to Machine Learning”. Enrollment: 3 students.	
	<i>Guest lectures:</i>	
	INSC 667: Computational Neuroscience (Lecture on AI and machine learning) 2026	
	CHEG 867/848: Comp. & Data Sci. Soft Materials Innov. & Discovery 2022, 2024–2026 (2–3 Lectures on Machine Learning)	
	KAAP 667: Responsible Conduct Research (Discussion on AI) 2024,2025	
	EGGG 101: Introduction to Engineering (Interview on AI) 2025	
	CISC890: Computer Science: New Graduate Students 2019–2025	
	ELEG 305: Signals and Systems 5/2019	
	CPEG 457/657: Search and Data Mining 4/2019	
	CISC 483/683: Introduction to Data Mining 3/2019	
	University of Manchester , Manchester, UK	
	<i>Guest lectures</i> , Text Mining (COMP 61332) Spring 2018	
	University of Liverpool , Liverpool, UK	
	<i>Instructor</i> Eng. Skills (ELEC 171/172) Matlab (5 weeks) Spring 2016, Fall 2016	
	<i>Guest lecturer</i> Neural Networks (ELEC 320) (8 weeks) Spring 2015	
	University of Florida , Gainesville, Florida, USA	
	<i>Guest lecturer</i> , Brain Machine Interfaces (EEL 6935) Fall 2011, Fall 2013	
	<i>Teaching Assistant</i> , Microprocessor Applications (EEL 4744) Fall 2009, Spring 2010	
	University of Nebraska-Lincoln (Omaha Campus) , Omaha, Nebraska, USA	
	<i>Undergraduate Teaching Assistant</i> , Microprocessor System Design (CEEN 4330) Spring 2009	
	<i>Undergraduate Teaching Assistant</i> , Digital Design and Interfacing (CEEN 3110) Fall 2008	

TRAINING IN PEDAGOGY AND MENTORING	– Culturally Aware Mentoring, Workshop and Introduction	
	CIMER, University of Wisconsin-Madison and University of Delaware	1/2021
	– Inclusive Teaching Professional Development Workshop Series, University of Delaware	
	College of Engineering Diversity Working Group	Spring/Fall 2019
	– Course Design Institute, University of Delaware	6/2019
	– Associate Fellow of The Higher Education Academy	3/2016
	awarded following “Teaching for Researchers” modules at University of Liverpool	

Service

UNIVERSITY SERVICE	– Co-Lead, Research Working Group, First State AI Institute	9/2025–8/2026
	– Reviewer, Intramural Seed Grant Program	2024
	– Reviewer, Pilot Project, UD Institute for Engineering Driven Health	2024, 2025
	– Neuroscience Planning Committee (Chairs: John Jeka/Anna Klintsova)	8/2019–3/2021
DATA SCIENCE INSTITUTE SERVICE	– Masters of Science in Data Science (academic advisor)	1/2020–present
	– Breakout session host, “Foundational AI”	4/7/2025
	<i>Data Science and DARWIN Symposium</i>	
	– Faculty Advisor, Data Science Community Hour	1/2021–5/2021
	– Session Chair, “DARWIN for Physics, Engineering, and Computer Science”	2/12/2021
	<i>DARWIN Computing Symposium</i>	
	– DSI Representative, Technology & Data Analytics Career Meetup	3/4/2020
COLLEGE OF ENGINEERING SERVICE	– Member, Data Science Symposium Planning Committee	4/2019–11/2019
	(Chairs: Greg Dobler & Zachary Collier)	
	– Mastering Data Science and Statistical Analysis Information Session	2 × in 2019
COLLEGE OF ENGINEERING SERVICE	– Jr. Faculty Advisory Council	Spring 2024
	– Presenter, COE NSF GRFP Workshop	9/2/2021, 8/29/2022, 8/30/2023
	– AI Symposium Committee	6/29/2023–9/26/2023
	– Exploring Intellectual Neighborhoods	7/12/2023
	– Mentor, COE NSF GRFP Coaching Program	2021
ECE DEPARTMENT SERVICE	– Undergraduate Committee (Chair: F. Kiamilev)	2025–present
	– Lead, AI Minor Curriculum	2025–2026
	– Faculty Search Committee, Artificial Intelligence (Chair: Gonzalo Arce)	2025–2026
	– Chair, ECE AI Engineering BS Program Ad-hoc Committee	2024–2025
	– Representative, Admitted Student Day	2 × 2026
	– Representative, Blue & Golden Saturdays	3 × 2019, 2 × 2020, 2 × 2021, 2 × 2025
	– Representative, COE Open House	2025
	– Faculty Search Committee, Artificial Intelligence (Chair: Fouad Kiamilev)	2024–2025
	– Presenter, Wednesday Tech Forum	1 × 2025
	– Secondary Appointments Committee (Chair: Keith Goossen)	2023–2025
	– Representative, Delaware Decision Days	2 × 2019, 3 × 2021, 1 × 2025
	– Faculty Search Committee, AI for Communications/Cybersecurity	2023–2024
	(Chair: Javier Garcia-Frias)	
	– Coordinator, ECE Seminar Series	Fall 2020, Spring 2022, Spring 2023
	– UD IEEE Student Chapter Branch Counselor	5/2019–8/2022
	– Member, ECE Activities Committee (Chair: Vishal Saxena)	9/2021–5/2022
	– Representative, Department of Energy Virtual Recruiting Event	2022
	– Member, ECE Strategic Planning Committee (Chair: Jamie Phillips)	9/2020–5/2021
	– Member, ECE Areas Ad-hoc Committee (Chair: Kenneth Barner)	Fall 2019

CIS	– CIS Representative, Executive Committee, MSDS Program	9/2021–5/2022
DEPARTMENT	– Faculty Search Committee, Computer & Information Sciences	2019–2020
SERVICE	(Chair: Chien-Chung Shen; search resulted in 3 tenure-track faculty hires.)	

PH.D. DISSERTATION COMMITTEE MEMBERSHIP AT UD:

41. Zhanhao “Charlie” Zhang, Electrical and Computer Engineering	Ph.D. defense, 6/2026
40. Abdullah Alrushud, Electrical and Computer Engineering	Ph.D. defense, 6/2026
39. Stephen Kronenberger, Chemical and Biomolecular Eng.	Ph.D. defense, 6/2026
38. Josiane Mukahirwa, Neuroscience	Ph.D. proposal, 6/2026
37. Andres Ramirez-Jaime, Electrical and Computer Engineering	Ph.D. defense, 6/2026
36. Noaman Mehmood, Electrical and Computer Engineering	Ph.D. defense, 5/2026
35. GilHwan Kim, Mechanical Engineering	Ph.D. defense, 4/2026
34. Milad Markhali, Bioinformatics Data Science	Ph.D. candidacy, 4/2026
33. Jesus Orozco, Mechanical Engineering	Ph.D. defense, 3/2025
32. Daniela Martin, Computer and Information Sciences	Ph.D. proposal, 12/2025
31. Mohammad Baksh, Electrical and Computer Engineering	Ph.D. proposal, 12/2025
30. Sayeh Rezaee, Electrical and Computer Engineering	Ph.D. proposal, 8/2025
29. Lars Folkerts, Electrical and Computer Engineering	Ph.D. defense, 12/2025
26. Abdalrahman Hmod Alblwi, Electrical and Computer Engineering	Ph.D. defense, 4/2025
25. Ke Ma, Materials Science and Engineering	Ph.D. defense, 10/2025
24. Rachel Viger, Electrical and Computer Engineering	Ph.D. defense, 3/2025
23. Ashuta Bhattarai, Computer and Information Sciences	Ph.D. defense, 2/2025
22. Vishruta Yawatkar, Bioinformatics Data Science	Ph.D. proposal, 2/2025
21. Yue Zhang, Electrical and Computer Engineering	Ph.D. defense, 10/2025
20. Olamide “Lamal” Ayodele, Financial Services Analytics	Ph.D. proposal, 12/2024
19. Ghazaleh Zehtab, Financial Services Analytics	Ph.D. proposal, 12/2024
18. Samet Bayram, Electrical and Computer Engineering	Ph.D. proposal, 12/2024
17. Kyle Regan, Bioinformatics Data Science	Ph.D. proposal, 9/2024
16. Raphael Poulain, Computer and Information Sciences	Ph.D. defense, 9/2024
15. Cameron Ibrahim, Computer and Information Sciences	Ph.D. proposal, 9/2024
14. Zahra Vahdat, Electrical and Computer Engineering	Ph.D. defense, 5/2024
13. Shizhao Lu, Chemical and Biomolecular Engineering	Ph.D. defense, 2/2024
12. Pasquale Zingo, Electrical and Computer Engineering	Ph.D. defense, 2/2024
11. Seyedalireza Khoshsirrat, Computer and Information Sciences	Ph.D. proposal, 1/2024
10. Mehak Gupta, Computer and Information Sciences	Ph.D. defense, 5/2023
9. Karelina Pena, Electrical and Computer Engineering	Ph.D. defense, 1/2023
8. Fanruo Meng, Electrical and Computer Engineering	Ph.D. defense, 1/2023
7. Xinjie “Ethan” Lan, Electrical and Computer Engineering	Ph.D. defense, 4/2022
6. Sergio Sepúlveda, Electrical and Computer Engineering	Ph.D. defense, 4/2022
5. Zhenzhu Zheng, Computer and Information Sciences	Ph.D. proposal, 11/2020

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| 4. Kevin Corder, Computer and Information Sciences | Ph.D. defense, 9/2025 |
| 3. Kuang Lu, Electrical and Computer Engineering | Ph.D. defense, 11/2020 |
| 2. Michael J. De Lucia, Electrical and Computer Engineering | Ph.D. defense, 3/2020 |
| 1. Alejandro Parada-Mayorga, Electrical and Computer Engineering | Ph.D. defense, 7/2019 |

M.S. THESES, COMMITTEE MEMBERSHIP AT UD:

- | | |
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| 3. Joseph Cristiano, Bioinformatics Data Science | M.S. defense, 7/2025 |
| 2. Yashwanth Tekumudi, Master of Science in Robotics | M.S. defense, 4/2024 |
| 1. Camryn Scully, Master of Science in Robotics | M.S. defense, 4/2024 |

SENIOR THESES, HONORS DEGREE, COMMITTEE MEMBERSHIP AT UD:

- | | |
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| 2. Kristina Holsapple, Computer and Information Sciences | 5/2023 |
| 1. Rebecca Clements, Biomedical Engineering | 5/2021 |

THESIS/PH.D.COMMITTEE MEMBERSHIP (EXTERNAL)

- | | |
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| Daniel Guillermo García Murillo, Automatic Engineering (Ingeniería Automática) | |
| Universidad Nacional de Colombia, Manizales, Colombia | D. Eng. defense, 9/2024 |

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| PROFESSIONAL INVOLVEMENT | – IEEE (Institute for Electrical and Electronics Engineers) | 2006–Present |
| | —Senior Member | 5/2025–Present |
| | —Delaware Bay Section, Student Activities Committee | 5/2019–2024 |
| | —University of Delaware Student Branch Counselor | 5/2019–8/2022 |
| | —Signal Processing Society | 2013–Present |
| | —Engineering in Medicine and Biology Society (EMBS) | 2010–Present |
| ACADEMIC SERVICE (REVIEWER) | – NSF Reviewer | 2021, 2022 |
| | – Pilot Grant Reviewer, Delaware INBRE | 2025 |
| | Journals: | |
| | – <i>Transactions on Machine Learning Research</i> (TMLR) | 2022–2026 |
| | – <i>IEEE Transactions on Neural Networks and Learning Systems</i> | 2015–2024 |
| | – <i>Neuroinformatics</i> | 2025 |
| | – <i>IEEE Transactions on Artificial Intelligence</i> | 2023 |
| | – <i>Proceedings of the National Academy of Sciences</i> (PNAS) | 2023 |
| | – <i>IEEE Signal Processing Letters</i> | 2022 |
| | – <i>IEEE Transactions on Automatic Control</i> | 2021 |
| | – <i>IEEE Transactions on Knowledge Data Engineering</i> | 2017–2020 |
| | – <i>IEEE Transactions on Signal Processing</i> | 2019, 2020 |
| | – <i>IEEE Access</i> | 2019 |
| | – <i>IEEE Transactions on Biomedical Engineering</i> | 2014, 2018 |
| | Conferences: | |
| | – <i>NeurIPS</i> | 2018–2025 |
| | – <i>WACV</i> | 2025,2026 |
| | – <i>ICML</i> | 2019,2021–2026 |
| | – <i>ICLR</i> | 2021–2026 |
| | – <i>ICASSP</i> | 2009,2018–2026 |
| | – <i>CVPR</i> | 2024–2025 |
| | – <i>AAAI</i> | 2018,2020–2022,2025,2026 |
| | – <i>AISTATS</i> | 2025,2026 |

	– <i>IJCNN</i>	2025
	– <i>MLSP</i>	2018–2025
	– <i>IEEE EMBS NER</i>	2013,2017,2019,2021,2023
	– <i>EMNLP</i>	2018
	Workshops:	
	– <i>Structured Data for Health (SD4H) Workshop at ICML</i>	2026
	– <i>ML4H 2025 Symposium</i>	2025
	– <i>NeurIPS Workshop on Times Series for Health (TS4H)</i>	2025
	– <i>NeurIPS Workshop on Distribution Shifts (DistShift)</i>	2022,2023
OUTREACH ACTIVITIES	– Exhibitor, “STEAM Week” (3rd, 4th, and 5th graders), 4Youth	4/8/2026
	– Co-Lead, “Catalyzing AI Challenge Teams in the Mid-Atlantic Region”	1/2026
	Recruited 3 teams and organized 4 sessions for the Presidential AI Challenge. One team as Delaware state champions.	
	– Presenter, Newark Charter School (Engineering Pathway at High School)	10/17/2025
	– Guest Speaker, “Looking under the hood at the AI engine”	9/9/2024
	Jenner’s Pond Retirement Community	
	– Mentor, Brain+AI Weekly Research Discussions (3 11th-grade students)	Summer 2024
	– Exhibitor, “STEAM Day at UD” (7th & 8th graders), Project Brain Light	4/9/2024
	– Instructor, “STEAM Day at Thomas Edison Charter School” (K–7th grades)	7/7/2023
	– Presenter, “45th Annual STEM Conference” (6–12th graders)	4/5/2023
	Delaware Technical Student Association	
	– Exhibitor, “STEAM Day at UD” (7th graders), Project Brain Light	4/4/2023
	– Module Preparation, “Read Someone’s Mind” (9 – 10 th graders), UD ECE	6/24/2022
	– Exhibitor, “STEAM Day at UD” (7th graders), Project Brain Light	6/7/2022
	– Project Judge, “UD GSG Hackathon on Misinformation and Cybersecurity”	4/25/2021
	– Presenter, “Engineering Your Tomorrow”, Sussex County (DE) STEM Alliance	2/2020
	– Presenter, “Measuring Electric Waves in the Brain”, Serviam Girls Academy	5/2019
	– Project Judge, FIRST LEGO League SE Pennsylvania Regional Championship	2/2019
	– Volunteer, Engineering Discovery Day, University of Delaware, AOE	10/2018